

# NATIONAL ROCKETRY CHAMPIONSHIP

## Flight Readiness Review

### UKC Astranova

Cara-Alexandra Hefer, Calum Lewis, Arda Aydinlar, Adam McGrath,  
Alex Dee, Theo Zilios, Lucian Willcock, Jack Shorey, Robert Parry



# Overview

| Task                          | Status   | Comments                                    |
|-------------------------------|----------|---------------------------------------------|
| Requirement Compliance        | Complete | There is one partial due to a late delivery |
| Launch Vehicle Manufacture    | Complete |                                             |
| OpenRocket Simulation         | Complete |                                             |
| Recovery System Assembly      | Complete |                                             |
| Recovery System Demonstration | Complete |                                             |
| Structural Tests              | Complete |                                             |

# Overview

| Task                          | Status   | Comments |
|-------------------------------|----------|----------|
| Launch Vehicle Decoration     | Complete |          |
| Rideshare Payload Manufacture | Complete |          |
| Rideshare Payload Code        | Complete |          |
| Rideshare Payload Tests       | Complete |          |
| The Customer's Payload        | Complete |          |
| Competition Day Procedure     | Complete |          |

# Requirement Compliance

| Requirement ID           | Met? [Yes, No, Partially] | Rational                                                                         |
|--------------------------|---------------------------|----------------------------------------------------------------------------------|
| NRC MRD-MIS-001   Rev: A | Yes                       | The motor chosen and design of the rocket theoretically should reach this apogee |
| NRC MRD-MIS-002   Rev: A | Yes                       | Dual-deployment is not being used                                                |
| NRC MRD-MIS-003   Rev: A | Yes                       | Rocket will be unable to reach this height                                       |
| NRC MRD-MIS-004   Rev: A | Yes                       | Motor was picked from provided list that complied with these guidelines          |
| NRC MRD-MIS-005   Rev: A | Yes                       | To be supplied by UKSED on the day                                               |
| NRC MRD-MIS-006   Rev: A | Yes                       | Confirmation email has been received                                             |
| NRC MRD-MIS-007   Rev: A | Yes                       | The RPD weighs 102g                                                              |
| NRC MRD-MIS-008   Rev: A | Yes                       | The rocket won't reach further than 750m                                         |

# Requirement Compliance

| Requirement ID           | Met? [Yes, No, Partially] | Rational                                                                                           |
|--------------------------|---------------------------|----------------------------------------------------------------------------------------------------|
| NRC MRD-CPD-001   Rev: B | Yes                       | A 5V 4A buck converter with an XT30 connector shall be used to power the CPD                       |
| NRC MRD-CPD-002   Rev: A | Yes                       | See NRC TSN-ESS-001   Rev: A                                                                       |
| NRC MRD-CPD-003   Rev: A | Yes                       | 18650CA-2S-3J battery has a max discharge current of 2.2A CPD has current draw of $7.5W/5V = 1.5A$ |
| NRC MRD-CPD-004   Rev: B | Yes                       | The buck converter used for powering the CPD as an XT30 connector                                  |
| NRC MRD-CPD-005   Rev: A | Yes                       | AstraNova will provide an sd card for the CPD                                                      |
| NRC MRD-CPD-006   Rev: B | Yes                       | The avionics sled has a a launch vehicale adapter mounted at the top for the CPD                   |
| NRC MRD-CPD-007   Rev: B | Yes                       | As per the model the the avionics sled has an diameter of 50mm and clearance of over 50mm          |



UKSADS  
COMPETITIONS



# Requirement Compliance

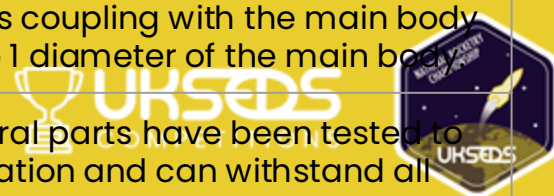
| Requirement ID           | Met? [Yes, No, Partially] | Rational                                                                                          |
|--------------------------|---------------------------|---------------------------------------------------------------------------------------------------|
| NRC MRD-CPD-008   Rev: B | Yes                       | As per the 3d assembly, the customer payload It is mounted concentrically                         |
| NRC MRD-CPD-009   Rev: B | Yes                       | Holes have been drilled into the side of the rocket for external access                           |
| NRC MRD-CPD-010   Rev: A | Yes                       | The entire avionic Sled is unpressurized                                                          |
| NRC MRD-CPD-011   Rev: A | Yes                       | AstraNova has m3 bolts to for mounting the CPD                                                    |
| NRC MRD-RPD-001   Rev: A | Yes                       | Custom PCB designed in Fusion360, manufactured and populated in-house                             |
| NRC MRD-RPD-002   Rev: A | Yes                       | BMP280 barometer, bench tested, altitude reading verified by walking up and down flight of stairs |
| NRC MRD-RPD-003   Rev: A | Yes                       | Python script generates plot from SD card data<br>Demonstrated in bench test video.               |

# Requirement Compliance

| Requirement ID           | Met? [Yes, No, Partially] | Rational                                                                                                                                                                                                                                                                                                 |
|--------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NRC MRD-RPD-004   Rev: A | Yes                       | Code calculates and logs apogee value to SD card and OLED display.                                                                                                                                                                                                                                       |
| NRC MRD-RPD-005   Rev: A | Yes                       | Silkscreen is on rideshare PCB(example photo in following slides)                                                                                                                                                                                                                                        |
| NRC MRD-RPD-006   Rev: A | Yes                       | 10-DOF IMU logs acceleration, gyro, mag data throughout flight.                                                                                                                                                                                                                                          |
| NRC TSN-MOR-001   Rev: A | Yes                       | Mount has been designed to maximise strength and minimise weight. Material selected has a sufficient heat tolerance. Will fit tightly to the top , secured with screws to the body – transferring thrust from the motor to the rocket. Rear clamps also fit tightly and secured with screws to the body. |
| NRC TSN-MOR-002   Rev: A | Yes                       | Has been tested and can withstand the force                                                                                                                                                                                                                                                              |
| NRC TSN-MOR-003   Rev: A | Yes                       | Has been tested and can withstand the force                                                                                                                                                                                                                                                              |
| NRC TSN-SMN-001   Rev: A | YES                       | See openrocket design                                                                                                                                                                                                                                                                                    |

# Requirement Compliance

| Requirement ID           | Met? [Yes, No, Partially] | Rational                                                                                                                      |
|--------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| NRC TSN-SMN-003   Rev: A | YES                       | See openrocket design                                                                                                         |
| NRC TSN-SAR-001   Rev: A | YES                       | Fins have been fixed in line with the longitudinal axis of the rocket                                                         |
| NRC TSN-SAR-002   Rev: A | YES                       | Fin designs where tested to specification and found to be able to withstand high levels of strain                             |
| NRC TSN-SAR-003   Rev: A | YES                       | See above                                                                                                                     |
| NRC TSN-SAR-004   Rev: A | YES                       | Main body was tested with various weights in excess of 2 kg and found to be resilient to high lateral and longitudinal forces |
| NRC TSN-SAR-005   Rev: A | YES                       | Nose cone's coupling with the main body is equal to 1 diameter of the main body                                               |
| NRC TSN-SAR-006   Rev: A | YES                       | All structural parts have been tested to specification and can withstand all expected loads                                   |



# Requirement Compliance

| Requirement ID           | Met? [Yes, No, Partially] | Rational                                                                                                                          |
|--------------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| NRC TSN-SAR-007   Rev: A | Partially                 | Rail buttons have been ordered however have not yet arrived                                                                       |
| NRC TSN-SAR-008   Rev: A | YES                       | FEA analysis was not used for parts testing                                                                                       |
| NRC TSN-ESS-001   Rev: A | Yes                       | Main battery 2S 2200mAh at ~60mA (rideshare) + customer payload at 1.5A. Total ~1.56A draw. 2200mAh at 7.8W = 50 minutes runtime. |
| NRC TSN-ESS-002   Rev: A | Yes                       | Avionics sled can be removed by unscrewing six screws. Allows removal and arming in under 5 minutes                               |
| NRC TSN-ESS-003   Rev: A | Yes                       | No single core wires have been used                                                                                               |
| NRC TSN-ESS-004   Rev: A | Yes                       | The team conforms to ofcom's IR 2030 RF ragulation as no radio has been used                                                      |
| NRC TSN-RRS-001   Rev: A | Yes                       | The vehicle will have a recovery system                                                                                           |



UKSDS  
COMPETITIONS



# Requirement Compliance

| Requirement ID           | Met? [Yes, No, Partially] | Rational                                                                                          |
|--------------------------|---------------------------|---------------------------------------------------------------------------------------------------|
| NRC TSN-RRS-002   Rev: A | Yes                       | The speed of the vehicle will be less than 15m/s                                                  |
| NRC TSN-RRS-003   Rev: A | Yes                       | The vehicle will not drift more than one kilometre                                                |
| NRC TSN-RRS-004   Rev: A | Yes                       | The flight computer used is UKRA qualified                                                        |
| NRC TSN-RRS-005   Rev: A | Yes                       | Flight controller uses separate battery and therefore is entirely isolated from Payload's Battery |
| NRC TSN-RRS-006   Rev: A | Yes                       | Tilt switches have not been used                                                                  |
| NRC TSN-RRS-007   Rev: A | Yes                       | The flight computer has an externally disarming mechanism                                         |
| NRC TSN-RRS-008   Rev: A | Yes                       | Eye bolts with screws have been used                                                              |

# Requirement Compliance

| Requirement ID           | Met? [Yes, No, Partially] | Rational                                     |
|--------------------------|---------------------------|----------------------------------------------|
| NRC TSN-RRS-007   Rev: A | Yes                       | The shock cords are above the minimum length |
| NRC TSN-RRS-008   Rev: A | Yes                       | The shock cords are made of kevlar           |
| NRC TSN-ARY-001   Rev: A | Yes                       | Recovery conforms to CAA UAS law             |
| NRC TSN-ARY-002   Rev: A | Yes                       | Recovery conforms to UKRA safety code        |
| NRC TSN-ARY-003   Rev: A | Yes                       | We are not using active stabilisation        |

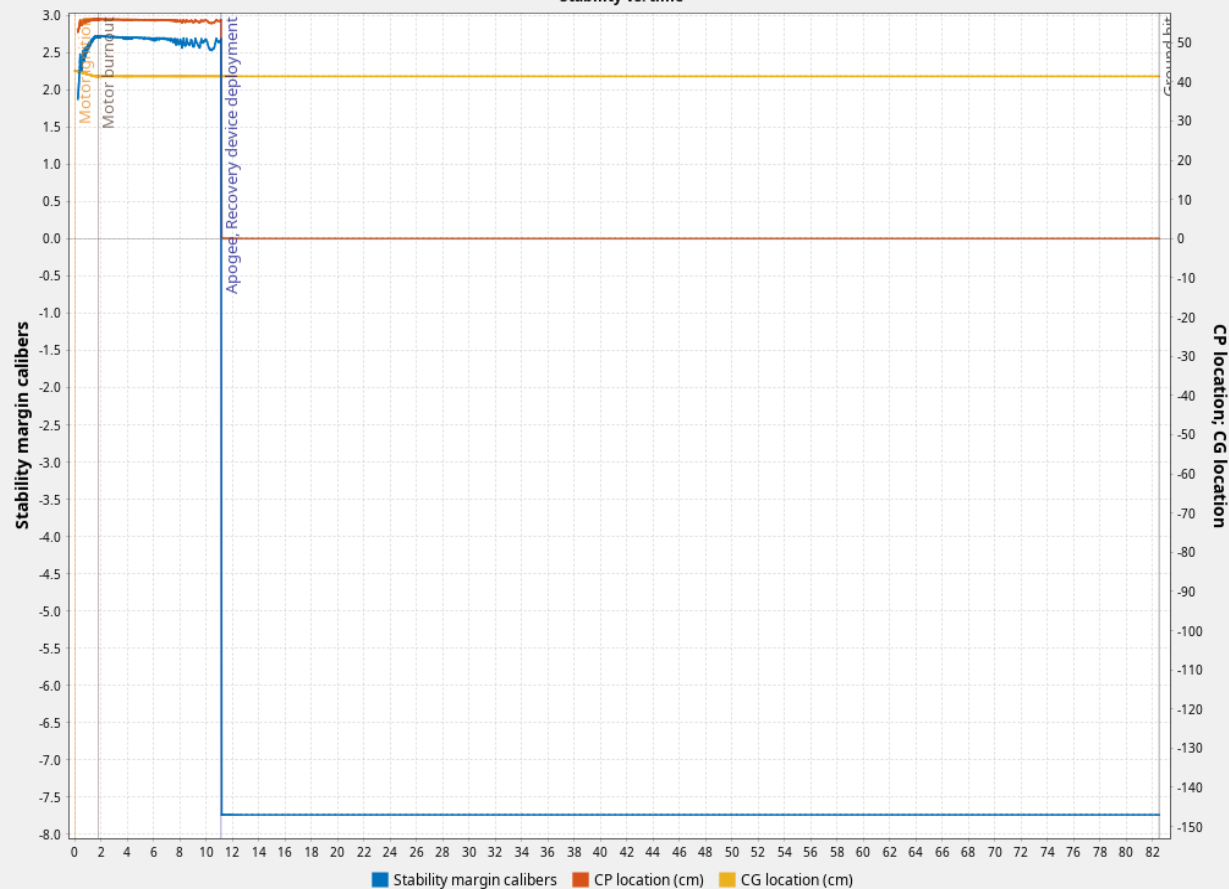


# Launch Vehicle Manufacture



## 3.5 m/s (Average)

Stability vs. time



## OpenRocket Simulation



Nose Cone configuration

Component name: **Nose Cone**

Custom

Parts Library

General

Shoulder

Override

Appearance

Comment

Nose cone shape: **Ogive**

Shape parameter: **1**

Length: **17** cm

Base diameter: **5.35** cm

Wall thickness: **0.3** cm

Material: **PVC (1.39 g/cm³)**

Component finish: **Mirror surface (0 µm)**

Set for all

Flip to tail cone

Cancel

OK

An ogive nose cone has a profile that is a segment of a circle. The shape parameter value 1 produces a tangent ogive, which has a smooth transition to the body tube, values less than 1 produce **semit ogives**.

Component mass: 101 g (overridden to 42.3 g)

Mass Component configuration

Component name: **Sled**

Custom

Parts Library

General

Radial position

Override

Appearance

Comment

Mass type: **Mass Component**

Placement: **Position relative to: Top of the parent component**

plus **6.3** cm

Mass: **46** g

Approximate density: **0.081** g/cm³

Length: **29** cm

Diameter: **5** cm

Automatic

Cancel

OK

Component mass: 46 g

Mass Component configuration

Component name: **Battery**

Custom

Parts Library

General

Radial position

Override

Appearance

Comment

Mass type: **Mass Component**

Placement: **Position relative to: Top of the parent component**

plus **22.5** cm

Mass: **102** g

Approximate density: **1.35** g/cm³

Length: **7** cm

Diameter: **3.7** cm

Automatic

Cancel

OK

Component mass: 102 g

Trapezoidal Fin Set configuration

Component name: **Trapezoidal Fin Set**

Custom

Parts Library

General

Fin tabs

Override

Appearance

Comment

Number of fins: **4**

Placement: **Position relative to: Bottom of the parent component**

plus **1** cm

Fin chord: **6** cm

Root chord: **5.7** cm

Tip chord: **5.7** cm

Height: **7** cm

Swamp length: **2.84** cm

Swamp angle: **38.7** °

Fin cross section: **Rounded**

Thickness: **0.3** cm

Material: **Polystyrene (1.05 g/cm³)**

Component finish: **Regular paint (60 µm)**

Set for all

Root fillet: **0** cm

Fill material: **Superior (DPS) (0.6 g/cm³)**

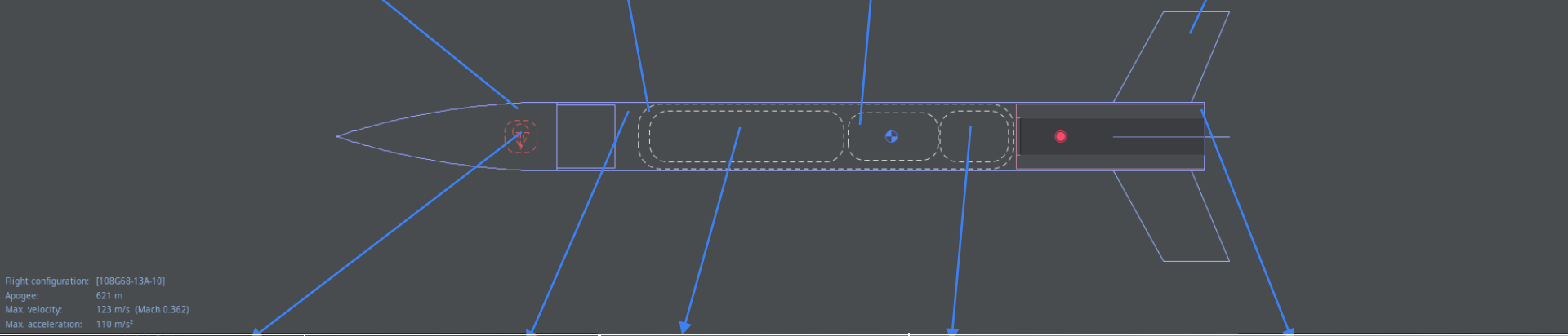
Cancel

OK

Component mass: 26.1 g (overridden to 27.8 g)

CDR design better  
 Length 68.9 cm, max. diameter 5.35 cm  
 Mass with no motors 613 g  
 Mass with motors 769 g

Stability: 2.44 cal / 19 %  
 CG: 42.8 cm  
 CP: 55.9 cm  
 at M=0.300



Flight configuration: [108G68-13A-10]  
 Apogee: 621 m  
 Max. velocity: 123 m/s (Mach 0.362)  
 Max. acceleration: 110 m/s²

Parachute configuration

Component name: **Parachute**

Custom

Parts Library

General

Radial position

Override

Appearance

Comment

Canopy: **50** cm

Placement: **Position relative to: Top of the parent component**

plus **13** cm

Material: **Polystyrene (1.05 g/cm³)**

Packed length: **2.5** cm

Packed diameter: **2.5** cm

Drag coefficient  $C_d$ : **0.800**

Should lines: **0**

Deployment: **plus **0** seconds**

Line length: **150** cm

Material: **Naval thread 800 (1.7 mm, 2644 wt (0.962 g/m))**

Altitude: **200** m

Cancel

OK

Component mass: 13.8 g (overridden to 15.8 g)

Body Tube configuration

Component name: **Body Tube**

Custom

Parts Library

General

Motor

Override

Appearance

Comment

Length: **50** cm

Outer diameter: **5.35** cm

Inner diameter: **5** cm

Wall thickness: **0.175** cm

Material: **Cardboard (0.68 g/cm³)**

Component finish: **Mirror surface (0 µm)**

Set for all

Automatic

Cancel

OK

Component mass: 96.7 g (overridden to 131 g)

Mass Component configuration

Component name: **Slider**

Custom

Parts Library

General

Radial position

Override

Appearance

Comment

Mass type: **Mass Component**

Placement: **Position relative to: Top of the parent component**

plus **7.2** cm

Mass: **100** g

Approximate density: **0.531** g/cm³

Length: **15** cm

Diameter: **4** cm

Automatic

Cancel

OK

Component mass: 100 g

Mass Component configuration

Component name: **Custom**

Custom

Parts Library

General

Radial position

Override

Appearance

Comment

Mass type: **Mass Component**

Placement: **Position relative to: Bottom of the parent component**

plus **15.1** cm

Mass: **75** g

Approximate density: **1.13** g/cm³

Length: **5.3** cm

Diameter: **4** cm

Automatic

Cancel

OK

Component mass: 75 g

Inner Tube configuration

Component name: **Inner Tube**

Custom

Parts Library

General

Motor

Cluster

Radial position

Override

Appearance

Comment

Length: **14.5** cm

Placement: **Position relative to: Bottom of the parent component**

plus **0** cm

Outer diameter: **5** cm

Inner diameter: **2.9** cm

Wall thickness: **1.05** cm

Material: **PLA - 100% infill (1.25 g/cm³)**

Component finish: **Regular paint (60 µm)**

Set for all

Cancel

OK

Component mass: 236 g (overridden to 32.9 g)

# Evidence of Component Mass



Structural components -205.67g



Sled - 46g



Recovery system components – 76.53g



Batteries -  
101.75g



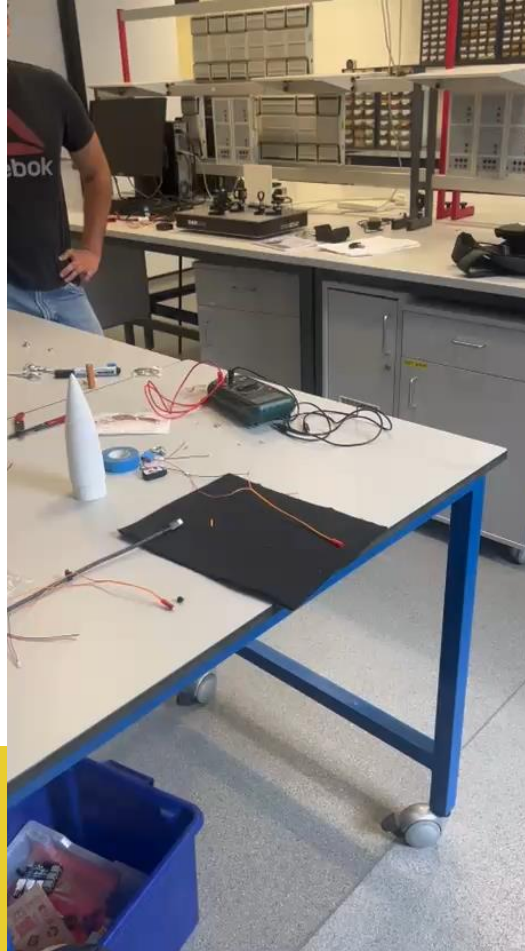
Screws (x15) -  
4.62g

# Recovery System Assembly



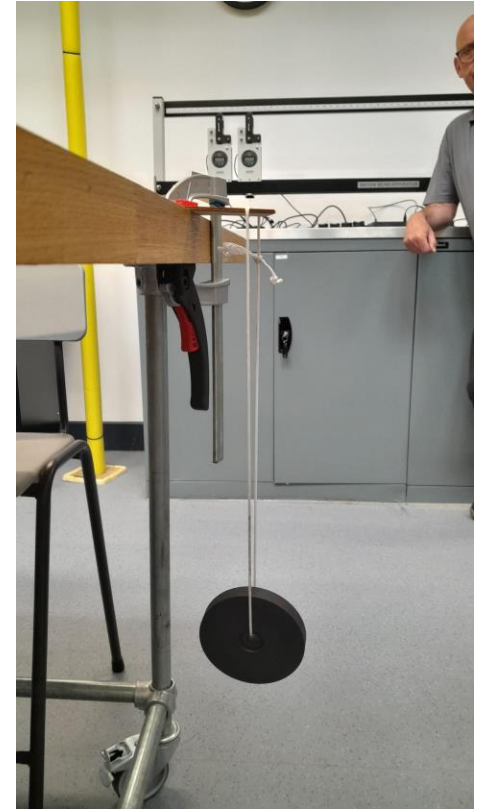
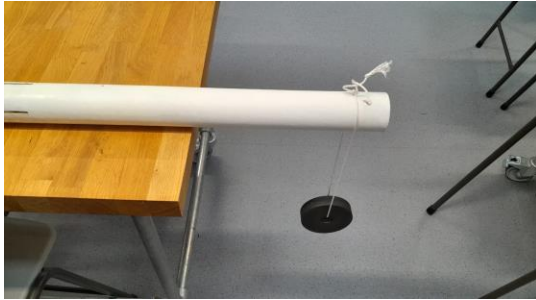
The video covers the following requirements:  
NRC RS-TSN-RRS-001  
NRC RS-TSN-RRS-004  
NRC RS-TSN-RRS-005  
NRC RS-TSN-RRS-006  
NRC RS-TSN-RRS-007  
NRC RS-TSN-RRS-008

# Recovery System Demonstration



# Structural Tests

Each structural component was tested using weights to simulate the longitudinal forces (and lateral for the fins) experienced during flight. The mass used was 800g for all components except for the main body, which was tested using 1200g. All parts passed their required stability requirements (NRC TSN – SAR – 001 -> 006)

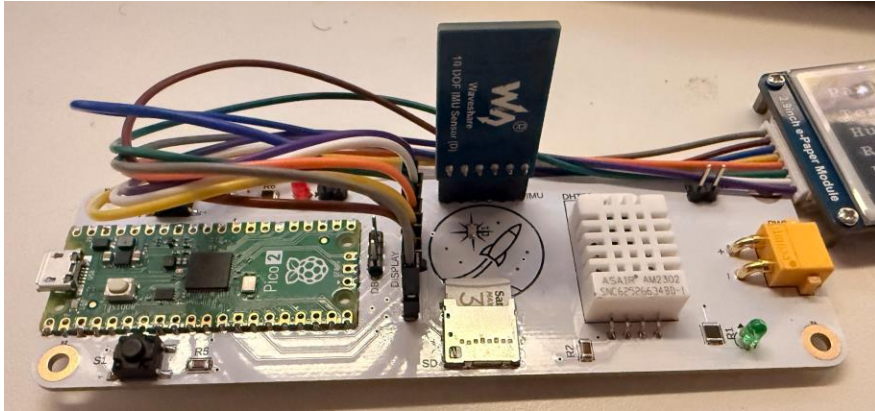




# Launch Vehicle Decoration

# Rideshare Payload Manufacture

NRC RS-MRD-RPD-005 | Rev: A  
NRC RS-MRD-RPD-001 | Rev: A



# Rideshare Payload Manufacture

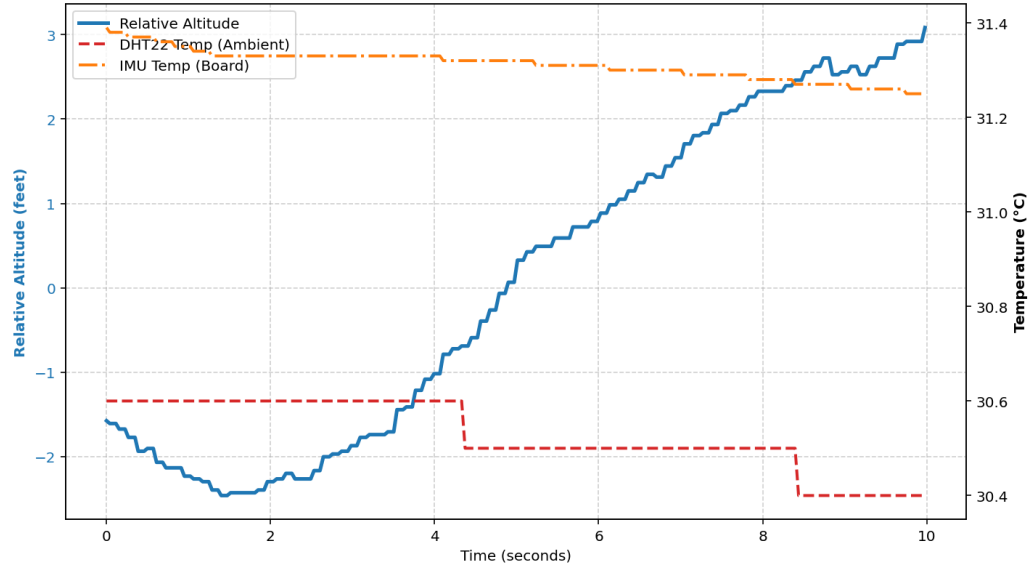


NRC MRD-MIS-007 | Rev: A

# Rideshare Payload Code

NRC RS-MRD-RPD-003 | Rev: A

Rocket Flight Data Analysis



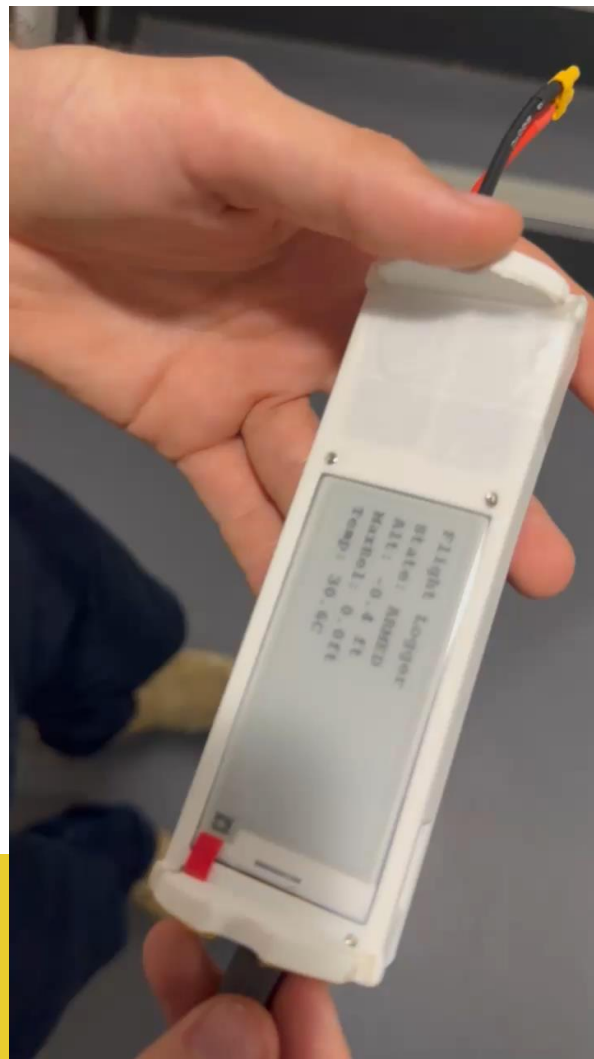
Example code of altitude time graph in feet, data taken from sd card and plotted using python matplotlib

# Rideshare Payload Code

NRC RS-MRD-RPD-006 | Rev: A

NRC RS-MRD-RPD-004 | Rev: A

NRC RS-MRD-RPD-002 | Rev: A



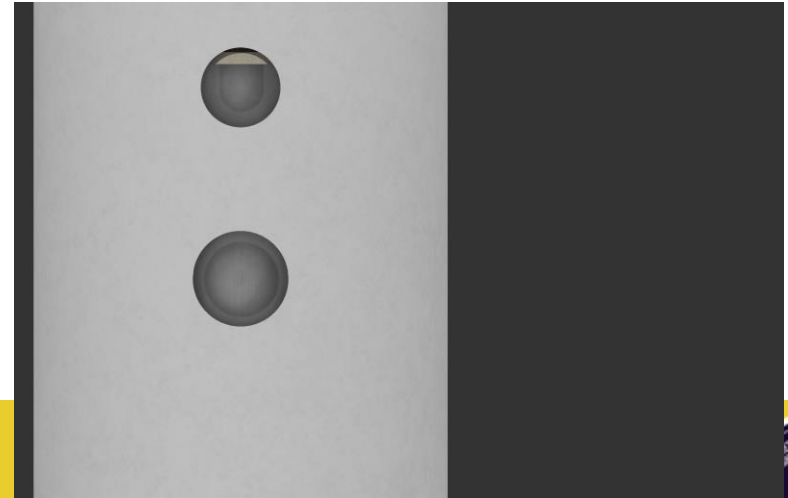
# Rideshare Payload Tests



# The Customer's Payload



NRC MRD-CPD-011 | Rev: A  
NRC RS-MRD-CPD-010 | Rev: A  
NRC RS-MRD-CPD-009 | Rev: A  
NRC RS-MRD-CPD-006 | Rev: A

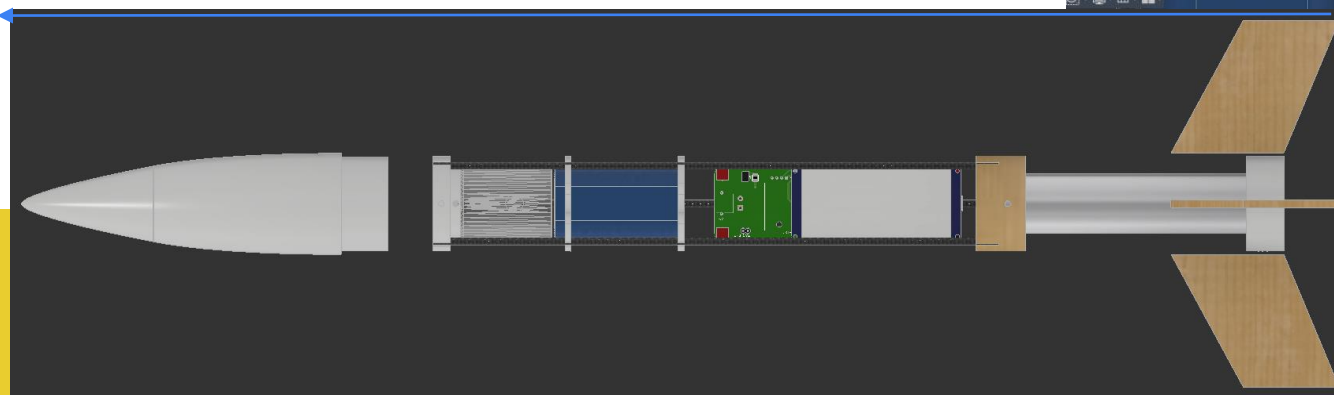
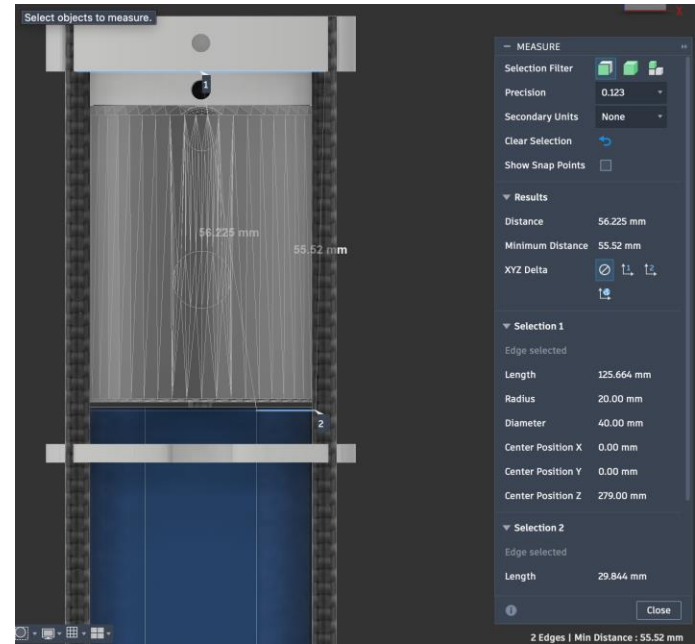


# The Customer's Payload

As per the design, the CPD has a thickened launch vehicle adapter in order to protect from any explosives and is over 20cm from the rocket motor, ensuring adequate protection.

NRC RS-MRD-CPD-008 | Rev: A

NRC RS-MRD-CPD-007 | Rev: A



# The Customer's Payload

DC-DC Buck Mini Module

5~36V wide voltage input, configurable 5V or 3.3V output, up to 4A output current



NRC RS-MRD-CPD-004 | Rev: A  
NRC RS-MRD-CPD-001 | Rev: A  
NRC RS-MRD-CPD-005 | Rev: A



# Competition Day Procedure

## Day before:

- Collect t-shirts
- Final electronics test
- Check all necessary parts are packed
- Any other extra tests to be done
- Assemble what can be assembled already
- Ensure transport
- Everyone is informed of times/plan

## Day of launch:

- Final checks
- Head count
- Leave on time for the venue
- Arrive at venue
- Register
- Final final check of rocket
- All assembly that could not be previously done to be completed

## At launch site:

- Recovery to check flight computer
- Avionics to set up rideshare and customer payload and make sure both are connected to the battery
- Motor checked to be fitted properly
- Structures final check on structural stability and ensure screws tightened
- Some encouraging words to the rocket
- Retreat to a safe distance

## After launch:

- Pack everyone into vehicle and head home with a hopefully mostly intact rocket!



# End Slide

